

Dr. Md. Saidur Rahman

36, Hull Street, Newcastle upon Tyne

NE4 6PU, UK

E-mail: msrahman@fwt.ku.ac.bd, Cell Phone: +447505450363

CURRICULUM VITAE

Date of Birth: September 01, 1984

Place of Birth: Jashore, Bangladesh

Citizenship: Bangladeshi

Khulna university profile: https://discipline.ku.ac.bd/fwt/faculty/ranju_fwt

Durham University profile: <https://www.durham.ac.uk/staff/md-s-rahman/#overview>

LinkedIn: <http://bd.linkedin.com/in/mdsrahman>

ORCID ID: <https://orcid.org/0000-0001-6849-4105>

Scopus ID: <https://www.scopus.com/authid/detail.uri?authorId=57217464411>

Key areas: mangrove ecology and management, Remote sensing and GIS, inventory and survey, carbon inventory, Ecosystem services, nature-based solutions.

PROFESSIONAL EXPERIENCES

Period	Designation	Institution	Responsibilities
09/2024- On going	Post Doctoral Research Associate	Durham University, UK	Developing an online Dashboard and/or application in the Google Earth Engine cloud computing environment to assist with routine monitoring of mangrove forest dynamics and coastal zone carbon stock changes in Suriname and Guyana.
11/2022- ongoing	Professor	Forestry and Wood Technology Discipline, Khulna University, Bangladesh	1. Teaching and examining undergraduate level students. 2. Supervising undergraduate and master's student for their project thesis. 3. Assess, review and evaluate student activities and progress. 4. Conducting substantive research into complex problems, ideas, concepts or theories and applying appropriate methodologies.
11/2016- 11/2022	Associate Professor	Forestry and Wood Technology Discipline, Khulna University, Bangladesh	As above

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11/2011 – 11/2016	Assistant Professor	Forestry and Wood Technology Discipline, Khulna University, Bangladesh	As above
11/2009 – 11/2011	Lecturer	Forestry and Wood Technology Discipline, Khulna University, Bangladesh	As above
07/2007 – 06/2008	Research Assistant	Forestry and Wood Technology Discipline, Khulna University, Bangladesh	1. Conducting field inventory in the Sundarbans mangrove forest 2. Data analysis and report preparation 3. Manuscript preparation

EDUCATION

10/2017- 11/2022	Doctor of Philosophy in Geography, Durham University UK. Research topic: Mapping above- and below-ground carbon stocks in the Sundarbans mangrove forest. Supervisors: Professor Daniel N M Donoghue, Professor Louise J Bracken
09/2013 -08/2015	Master of Science (Forest and Nature Conservation) specialization in ecology, Wageningen University and Research Centre, The Netherlands. Thesis topic: Salinity drives growth dynamics of the mangrove tree <i>Sonneratia apetala</i> Buch. -Ham. in the Sundarbans, Bangladesh Supervisors: Professor Pieter A. Zuidema, Professor Ute Sass-Klaassen, Professor Md. Qumruzzaman Chowdhury.
03/2008 – 08/2009	Master of Science (Forestry) specialization in ecology and management, Khulna University, Bangladesh. Thesis topic: Estimation of above ground biomass of <i>Ceriops decandra</i> in the Sundarbans mangrove forest, Bangladesh Supervisor: Dr. Mahmood Hossain
04/2002 – 03/2007	Bachelor of Science (Honours) in Forestry. Khulna University, Bangladesh Thesis topic: Nutrients (N, P and K) dynamics associated with the leaf litter of two agroforestry tree species of Bangladesh. Supervisor: Professor Dr. Mahmood Hossain
07/1999-09/2001	Higher Secondary Certificate , Cantonment College, Jessore, Bangladesh
01/1997-06/1999	Secondary School Certificate , B.A.F. Shaheen School and College, Jessore, Bangladesh

JOURNAL ARTICLES

- **Rahman, M. S.,** Shanu, S. A. and Amin, M. N. 2025. Socio-ecological perspectives of Urban Green Spaces (UGSs) policies in Bangladesh. Landscape, Ecology and Infrastructure. In:

Mallick, F. H. (Ed). Next50 Collective Futures. The University Press Limited, Dhaka. Pp 326-332.0 <https://www.next50bangladesh.com/aboutnext50>

- Chowdhury, M. Q., Sarker, S. K., Marma, M., **Rahman, M. S.** and Datta, A. 2024. Climate and salinity together control above ground carbon accumulation in the Sundarbans mangrove ecosystem. *Ocean & Coastal Management*, 255, 107242. doi:<https://doi.org/10.1016/j.ocecoaman.2024.107242>
- Bhargava, R., **Rahman, M. S.**, Ghosh, T. and Friess, D. A. 2024. Variations in biophysical characteristics of mangroves along retreating and advancing shorelines. *Science of The Total Environment*, 171690. <https://doi.org/10.1016/j.scitotenv.2024.171690>
- Zaman, M. R., **Rahman, M. S.**, Ahmed, S., & Zuidema, P. A. (2023). What drives carbon stocks in a mangrove forest? The role of stand structure, species diversity and functional traits. *Estuarine, Coastal and Shelf Science*, 108556. <https://doi.org/10.1016/j.ecss.2023.108556>
- Salekin, S., Hossain, M.N., Alam, M.A., Limon, M. S. H. and **Rahman, M. S.** 2023. Inter-specific competition between seeds and seedlings of two confamilial tropical trees. *Community Ecology*. <https://doi.org/10.1007/s42974-023-00165-3>
- Ahmed, S., Kamruzzaman, M., **Rahman, M. S.**, Sakib, N., Azad, M. S., & Dey, T. (2022). Stand structure and carbon storage of a young mangrove plantation forest in coastal area of Bangladesh: the promise of a natural solution. **Nature-Based Solutions**, 100025.
- **Rahman, M.S.**, Donoghue, D.N.M., Bracken, L.J. and Mahmood, H. 2021. Biomass estimation in mangrove forests: a comparison of allometric models incorporating species and structural information. **Environmental Research Letters** 16, 124002. <https://doi.org/10.1088/1748-9326/ac31ee>
- **Rahman, M.S.**, Donoghue, D.N.M., Bracken, L.J., 2021. Is soil organic carbon underestimated in the largest mangrove forest ecosystems? Evidence from the Bangladesh Sundarbans. **CATENA**. 200, 105159. <https://doi.org/10.1016/j.catena.2021.105159>
- Sultana, N., Limon, S.H., **Rahman, M.S.**, Akther, A., Salekin, S., Meason, D.F., Bloomberg, M., 2021. Germination and growth responses to water stress of three agroforestry tree species from Bangladesh. **Environmental Challenges**, 100256. <https://doi.org/10.1016/j.envc.2021.100256>.
- **Rahman, M.S.**, Alam, M.A., Salekin, S., Belal, M.A.H. and Rahman, M. S. 2021. The COVID-19 pandemic: a threat for forest and wildlife conservation in Bangladesh? **Trees, Forests and People** 5, 100119. <https://doi.org/10.1016/j.tfp.2021.100119>
- Amin, M. N., Islam, S., **Rahman, M. S.** and Snigdha, S. S. 2021. Influence of humanitarian assistances on ecosystem services in Bangladesh coast. **International Journal of Forestry, Ecology and Environment**, 04(01), pp. 153-162. <https://doi.org/10.18801/ijfee.040121.17>

- **Rahman, M. S.**, Sass-Klassen, U., Zuidema, P. A., Chowdhury, M. Q. and Beeckman, H. Salinity drives growth dynamics of the mangrove tree *Sonneratia apetala* Buch.-Ham. in the Sundarbans, Bangladesh. **Dendrochronologia** 62, 125711. <https://doi.org/10.1016/j.dendro.2020.125711>.
- Mitu, M., Islam, M. A., **Rahman, M. S.** Feroz, M. S., Mollick, A. S. and Kabir, M. E. 2019. Pyrolysis kinetic study on waste particle residue from particle board industry. **Journal of the Indian Academy of Wood Science** 16 (1), 58-66. <https://doi.org/10.1007/s13196-019-00236-x>.
- Shanu, S. A. and **Rahman, M. S.** 2017. Exploring Cross-Sectoral Adaptation Challenges in the Coastal Areas of Bangladesh. In: Islam, Z., Shafie, H. and Mahmood, R. (eds.). Culture, Adaptation and Resilience: Essays on Climate Change Regime in South Asia. Bangladesh Climate Change Trust (BCCT) and Department of Anthropology, University of Dhaka, Bangladesh. pp: 201-215.
- Mahmood, H., Siddique, M. R. H., Abdullah, S. M. R., Saha, S., Ghosh, D. C., **Rahman, M. S.** and Limon, S. H. 2014. Nutrient Dynamics Associated with Leaching and Microbial Decomposition of Four Abundant Mangrove Species Leaf Litter of the Sundarbans, Bangladesh. **Wetlands** 34(3), 439–448. <https://doi.org/10.1007/s13157-013-0510-1>.
- Rahman, M. S. and **Rahman, M. S.** 2013. Ownership status of private woodlot owners with particular reference to Jessore District. **Journal of Bangladesh Academy of Agriculture** 5(1), 44-48.
- Mahmood, H., Siddique, M. R. H., **Rahman, M. S.**, Hossain, M. Z. and Hasan, M. M. 2011. Nutrient dynamics associated with leaf litter decomposition of three agroforestry tree species (*Azedarachta indica* A. Juss., *Dalbergia sissoo* Roxb., and *Melia azedarach* L.) of Bangladesh. **Journal of Forestry Research** 22(4), 577-582. <https://doi.org/10.1007/s11676-011-0175-7>.
- Mahmood, H., Limon, S. H., **Rahman, M. S.**, Azad, A. K., Islam, M. S. and Khairuzzaman, M. 2009. Nutrients (N, P and K) dynamics associated with the leaf litter of two agroforestry tree species of Bangladesh. **iForest-Biogeosciences and Forestry** 2(5), 183-186. <https://doi.org/10.3832/ifor0510-002>.

RESEARCH EXPERIENCES

Present

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- Project Director on “Mapping top-dying disease of the *Heritiera fomes* tree species in the Bangladesh Sundarbans” funded by Khulna University Research and Innovation Cell, Khulna University, Khulna, Bangladesh (BDT 6,50,000/-)
- Forestry expert on “*Mapping Bi-Directional Ecosystem Services of Nature-Based Solutions: A Case Study of Coastal Forests in Bangladesh*” in collaboration with International Centre for Climate Change and Development (ICCCAD), Bangladesh funded by the Bangladesh Forest Department (BDT 75,00,000/-).

Past

- Estimation of above ground biomass of *Ceriops decandra* and *Aegialitis rotundifolia* in the Sundarbans mangrove forest, Bangladesh
- Awarder fellowship for the project Nutrient dynamics associated with leaf litter leaching and decomposition of Sundri (*Heritiera fomes*) in the Sundarbans by the Ministry of Science and Information & Communication Technology, Government of the People’s Republic of Bangladesh, 2008.
- Project Director of research project on Effect of salinity on the germination and growth of Amur (*Amoora cucullata*) funded by research cell, Khulna University, Khulna, Bangladesh, 2009-10.
- Associate Investigator of research project on Home gardening for biodiversity conservation and livelihood options: an address towards climate change and poverty alleviation in Bangladesh- Funded by the Ministry of Science and Information & Communication Technology, Government of the People’s Republic of Bangladesh, 2010-2011.
- Associate Investigator of research project on Analyzing stand stratification, species diversity and structure of the Sundarbans mangrove forest: an ecological approach towards conservation of biodiversity- Funded by the Ministry of Science and Information & Communication Technology, Government of the People’s Republic of Bangladesh, 2010-2011.
- Co-Investigator of research project on Home gardening for biodiversity conservation and rural livelihood option: a case of Satkhira District funded by research cell, Khulna University, Khulna, Bangladesh.
- Project Director of research project on Effect of salinity on the germination and growth of Passur (*Xylocarpus mekongensis*) funded by University Grant Commission, Bangladesh, 2009-10.

- Co-Investigator of research project on an investigation of drought tolerance of selected agroforestry tree species through germination and growth performance: a livelihood adaptation option to climate change in Bangladesh funded by research cell, Khulna University, Khulna, Bangladesh.
- Associate investigator of Effect of herbicides and mechanical control treatments to manage invasive *Wedelia trilobata* Funded by Research Cell, Khulna University 2015.
- Project Director of Do radiofrequencies from mobile phone base stations affect the productivity of *Cocos nucifera*? Funded by Research Cell, Khulna University 2016.
- Associate investigator on “Mapping the impact of climate change on vegetation dynamics in the Sundarbans” in collaboration with Forestry and Wood Technology Discipline, Khulna University funded by Project of Administrative Convention Centre Development in Khulna City, Khulna, Bangladesh (BDT 10,00,000/_).

CONFERENCES

- Shanu, S. A. and **Rahman, M. S.**: Farmer’s Perceptions and Decisions: An Application of Adaptation Pathway in the Context of climate-vulnerable Bengal Delta, EGU General Assembly 2024, Vienna, Austria, 14–19 Apr 2024, EGU24-19921, <https://doi.org/10.5194/egusphere-egu24-19921>, 2024.
- Oral presentation in the **2nd International Conference on Environment** on Spatio-temporal dynamics in land use and land cover of the GBM delta in Bangladesh at Environmental Science Discipline, Khulna University, Khulna 9208, Bangladesh, 02-03 March 2024.
- Oral presentation in the **2nd International Conference on Environment** on Mapping top-dying disease of the *Heritiera fomes* tree species in the Bangladesh Sundarbans at Environmental Science Discipline, Khulna University, Khulna 9208, Bangladesh, 02-03 March 2024.
- Poster presentaiton in the **2nd International Conference on Environment** on Mapping spatial dynamics of urban green spaces in the Khulna city, Bangladesh at Environmental Science Discipline, Khulna University, Khulna 9208, Bangladesh, 02-03 March 2024.
- Oral presentation in the **UK National Earth Observation Conference’ 2022** on “Mapping above- and below-ground carbon stock in the Sundarbans mangrove forest, Bangladesh” at **National Space Centre**, Leicester on 6 - 8 September, 2022.
- Presented poster on “Biomass estimation in mangrove forest: a comparison of allometric models incorporating species and structural information” in the **XV World Forestry Congress**, Seoul, Republic of Korea on 02- 06 May’ 2022.

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- Presented research on “Biomass estimation in mangrove forest: a comparison of allometric models incorporating species and structural information” in the **20th Commonwealth forestry conferences** organised by the University of British Columbia, Canada on 16-18 August’ 2021.
- Presented poster on “Uncertainties in estimation of Spatial carbon dynamics in the Sundarbans mangrove forest” in the **5th international Mangrove Macro-benthos & Management meeting (MMM5)** (1-5 July’ 2019).
- Presented research on “Exploring Cross-Sectoral Adaptation Challenges in the Coastal Areas of Bangladesh” in the **international conference Anthropology, Adaptation and Resilience in Climate Change Regime** organised by Bangladesh Climate Change Trust (BCCT) and Department of Anthropology, University of Dhaka, Bangladesh (22-23 October’ 2016).
- Presented poster on “Growth dynamics of *Sonneratia apetala* from the Sundarbans is driven by salinity” in wood science symposium on **Wood science underpinning tropical forest ecology and management, Tervuren, Belgium** (May 26-29, 2015).

AWARDS, SCHOLARSHIPS AND PRIZES

- FoCAS Fellow (UNH and USFS, USA): Fellow in the Forestry and Climate Science Visiting Fellows Program (FoCaS-VEP) 2023 jointly organized by the University of New Hampshire, USA and USFS-IP’s COMPASS project in Bangladesh funded by USAID.
- Commonwealth PhD Scholarship for pursuing PhD in the Department of Geography, Durham University by Commonwealth Scholarship Commission in the UK (tuition fees, living allowances, research fees and travel allowances).
- Small project grant from Institute of Hazard, Risk and Resilience (IHRR), Durham University, UK (£2,000).
- Durham research grant by the Department of Geography (£500).
- Conference travel grant by Ustinov college, Durham University (£500).
- Netherlands Fellowship Programme (NFP) for MSC in forest and nature conservation in Wageningen University and Research, The Netherlands.

PROFESSIONAL AFFILIATIONS

- “Associate Fellow” of the UK Professional Standards Framework (UKPSF) for teaching and supporting learning in higher education.
- Former Member of British Ecological Society, the UK.
- General Member of the Institute of Foresters Bangladesh (IFB), Bangladesh (10716).

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- Life member of the Ecological Society of Bangladesh, Bangladesh (LM-324)

Trainings

- Conducted training on “Applications of Google Earthe Engine (GEE) in Forestry” to students and forestry professionals in Bangladesh in 2021, 2022 and 2023 funded by COMPASS program of USAID and USFS.

REFERENCES

- **Dr. Daniel N. M. Donoghue**
Professor, Department of Geography
Durham University
Stockton Road, Durham
DH1 3LE, UK
Phone: +44 (0) 1913341867
E-mail: danny.donoghue@durham.ac.uk
- **Dr. Louise J. Bracken**
Pro Vice-Chancellor (Research and Innovation)
Northumbria University
Sutherland Building
Newcastle-upon-Tyne
NE1 8ST, UK
Phone: +44 (0) 1912274004
E-mail: louise.bracken@northumbria.ac.uk
- **Dr. A. K. Fazlul Hoque**
Professor, Forestry and Wood Technology Discipline
Khulna University, Khulna-9208
Bangladesh
Phone: +8801715168551
E-mail: akf_hoque@yahoo.com